Elementare Gleichungen

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Inhaltsverzeichnis

1 Einführung

1.1 Aufgaben

I. Löse die folgenden Glleichungen:

1. Löse die Gleichung $2x + 4 = 18$.

$$\begin{aligned} 2x + 4 &= 18 && \big| -4 \\ 2x &= 14 && \big| \div 2 \\ x &= \mathbf{7} \end{aligned}$$

2. Löse die Gleichung $6x + 3 = 33$.

$$\begin{aligned} 6x + 3 &= 33 && \big| -3 \\ 6x &= 30 && \big| \div 6 \\ x &= \mathbf{5} \end{aligned}$$

3. Löse die Gleichung $5x + 8 = 13$.

$$\begin{aligned} 5x + 8 &= 13 && \big| -8 \\ 5x &= 5 && \big| \div 5 \\ x &= \mathbf{1} \end{aligned}$$

4. Löse die Gleichung $3x + 9 = 18$.

$$\begin{aligned} 3x + 9 &= 18 && \big| -9 \\ 3x &= 9 && \big| \div 3 \\ x &= \mathbf{3} \end{aligned}$$

5. Löse die Gleichung $6x + 9 = 45$.

$$\begin{aligned} 6x + 9 &= 45 && \big| -9 \\ 6x &= 36 && \big| \div 6 \\ x &= \mathbf{6} \end{aligned}$$

6. Löse die Gleichung $4x + 9 = 29$.

$$\begin{aligned} 4x + 9 &= 29 && \big| -9 \\ 4x &= 20 && \big| \div 4 \\ x &= \mathbf{5} \end{aligned}$$

7. Löse die Gleichung $7x + 4 = 25$.

$$\begin{aligned} 7x + 4 &= 25 && \big| -4 \\ 7x &= 21 && \big| \div 7 \\ x &= \mathbf{3} \end{aligned}$$

8. Löse die Gleichung $2x + 7 = 11$.

$$\begin{aligned} 2x + 7 &= 11 && \Big| -7 \\ 2x &= 4 && \Big| \div 2 \\ x &= \mathbf{2} \end{aligned}$$

9. Löse die Gleichung $4x + 8 = 36$.

$$\begin{aligned} 4x + 8 &= 36 && \Big| -8 \\ 4x &= 28 && \Big| \div 4 \\ x &= \mathbf{7} \end{aligned}$$

10. Löse die Gleichung $6x + 4 = 46$.

$$\begin{aligned} 6x + 4 &= 46 && \Big| -4 \\ 6x &= 42 && \Big| \div 6 \\ x &= \mathbf{7} \end{aligned}$$

2 Negative Zahlen

2.1 Aufgaben zu Kapitel 2.1 bis 2.3

I. Berechne die folgenden Summen:

1. $4 + (-3) = 4 - 3 = \mathbf{1}$
2. $5 + (-2) = 5 - 2 = \mathbf{3}$
3. $4 + (-6) = 4 - 6 = -\mathbf{2}$
4. $9 + (-7) = 9 - 7 = \mathbf{5}$
5. $8 + (-3) = 8 - 3 = \mathbf{5}$
6. $4 + (-17) = 4 - 17 = -\mathbf{13}$
7. $14 + (-5) = 14 - 5 = \mathbf{9}$
8. $1 + (-8) = 1 - 8 = -\mathbf{7}$
9. $9 + (-12) = 9 - 12 = -\mathbf{3}$
10. $7 + (-20) = 7 - 20 = -\mathbf{13}$
11. $23 + (-12) = 23 - 12 = \mathbf{11}$
12. $33 + (-21) = 33 - 21 = \mathbf{12}$
13. $14 + (-6) = 14 - 6 = \mathbf{8}$
14. $25 + (-30) = 25 - 30 = -\mathbf{5}$
15. $5 + (-9) = 5 - 9 = -\mathbf{4}$
16. $17 + (-8) = 17 - 8 = \mathbf{9}$

17. $11 + (-18) = 11 - 18 = \mathbf{-7}$
18. $17 + (-25) = 17 - 25 = \mathbf{-8}$
19. $27 + (-6) = 27 - 6 = \mathbf{21}$
20. $13 + (-43) = 13 - 43 = \mathbf{-30}$
21. $4 + (-1) = 4 - 1 = \mathbf{3}$
22. $-2 + 3 = \mathbf{1}$
23. $-2 + (-4) = -2 - 4 = \mathbf{-6}$
24. $3 + (-5) = 3 - 5 = \mathbf{-2}$
25. $-2 + 1 = \mathbf{-1}$
26. $-5 + (-3) = -5 - 3 = \mathbf{-8}$
27. $4 + (-5) = 4 - 5 = \mathbf{-1}$
28. $-3 + (-4) = -3 - 4 = \mathbf{-7}$
29. $-2 + (-7) = -2 - 7 = \mathbf{-9}$
30. $2 + (-8) = 2 - 8 = \mathbf{-6}$
31. $-1 + (-1) = -1 - 1 = \mathbf{-2}$
32. $4 + (-7) = 4 - 7 = \mathbf{-3}$
33. $-2 + (-3) = -2 - 3 = \mathbf{-5}$
34. $-3 + 5 = \mathbf{2}$
35. $-3 + (-4) = -3 - 4 = \mathbf{-7}$
36. $-1 + (-7) = -1 - 7 = \mathbf{-8}$
37. $-3 + 4 = \mathbf{1}$
38. $2 + (-6) = 2 - 6 = \mathbf{-4}$
39. $-3 + 2 = \mathbf{-1}$
40. $-2 + (-8) = -2 - 8 = \mathbf{-10}$
41. $2 + (-2) = 2 - 2 = \mathbf{0}$
42. $-1 + 1 = \mathbf{0}$
43. $-3 + 3 = \mathbf{0}$
44. $5 + (-5) = 5 - 5 = \mathbf{0}$
45. $-7 + 7 = \mathbf{0}$
46. $13 + (-13) = 13 - 13 = \mathbf{0}$
47. $-8 + 8 = \mathbf{0}$
48. $35 + (-35) = 35 - 35 = \mathbf{0}$

II. Addiere die benachbarten Zahlen und schreibe das Ergebnis darüber. Mithilfe der obersten Zahl kann man erkennen, ob man richtig gerechnet hat.

a)

				-6				
			-4		-2			
		-1		-3		1		
	1		-2		-1		2	
-2		3		-5		4		-2

b)

				-8				
			-1		-7			
		1		-2		-5		
	1,5		-0,5		-1,5		-3,5	
3,5		-2		1,5		-3		-0,5

2.2 Aufgaben zu Kapitel 2.4

I. Berechne die folgenden Differenzen:

1. $4 - 3 = \mathbf{1}$
2. $8 - 5 = \mathbf{3}$
3. $5 - 7 = -\mathbf{2}$
4. $10 - 7 = \mathbf{3}$
5. $6 - 19 = -\mathbf{13}$
6. $13 - 5 = \mathbf{8}$
7. $2 - 9 = -\mathbf{7}$
8. $11 - 14 = -\mathbf{3}$
9. $6 - 30 = -\mathbf{24}$
10. $34 - 23 = \mathbf{11}$
11. $15 - 7 = \mathbf{8}$
12. $36 - 41 = -\mathbf{5}$
13. $4 - 8 = -\mathbf{4}$
14. $12 - 17 = -\mathbf{5}$
15. $18 - 26 = -\mathbf{8}$
16. $26 - 5 = \mathbf{21}$

17. $15 - 45 = -\mathbf{30}$
18. $-9 - (-2) = -9 + 2 = -\mathbf{7}$
19. $3 - (-2) = 3 + 2 = \mathbf{5}$
20. $-4 - (-1) = -4 + 1 = -\mathbf{3}$
21. $-2 - 3 = -\mathbf{5}$
22. $-2 - 4 = -\mathbf{6}$
23. $3 - (-5) = 3 + 5 = \mathbf{8}$
24. $-2 - 1 = -\mathbf{3}$
25. $-5 - 3 = -\mathbf{8}$
26. $-3 - (-4) = -3 + 4 = \mathbf{1}$
27. $-2 - (-4) = -2 + 4 = \mathbf{2}$
28. $-1 - (-1) = -1 + 1 = \mathbf{0}$
29. $4 - (-7) = 4 + 7 = \mathbf{11}$
30. $-2 - (-3) = -2 + 3 = \mathbf{1}$
31. $-3 - 5 = -\mathbf{8}$
32. $-3 - (-4) = -3 + 4 = \mathbf{1}$
33. $-1 - 7 = -\mathbf{8}$
34. $-3 - 4 = -\mathbf{7}$
35. $2 - (-6) = 2 + 6 = \mathbf{8}$
36. $-3 - 2 = -\mathbf{5}$
37. $-2 - (-8) = -2 + 8 = \mathbf{6}$
38. $-2 - 2 = -\mathbf{4}$
39. $-1 - 1 = -\mathbf{2}$
40. $4 - (-4) = 4 + 4 = \mathbf{8}$
41. $7 - (-7) = 7 + 7 = \mathbf{14}$
42. $35 - (-35) = 35 + 35 = \mathbf{70}$
43. $3 - 5 = -\mathbf{2}$
44. $-4 - 6 = -\mathbf{10}$
45. $-9 - (-6) = -9 + 6 = -\mathbf{3}$
46. $-4 - (-8) = -4 + 8 = \mathbf{4}$
47. $5 - (-4) = 5 + 4 = \mathbf{9}$
48. $4 - 7 = -\mathbf{3}$

II. Subtrahiere die benachbarten Zahlen (linke Zahl minus rechte Zahl) und schreibe das Ergebnis darüber. Mithilfe der obersten Zahl kann man erkennen, ob man richtig gerechnet hat.

					-62				
				-30		32			
a)			-13		17		-15		
		-5		8		-9		6	
	-2		3		-5		4		-2

					32				
				17		-15			
b)			9		-8		7		
		5,5		-3,5		4,5		-2,5	
	3,5		-2		1,5		-3		-0,5

2.3 Aufgaben zu Kapitel 2.5

I. Berechne die folgenden Summen/Differenzen:

1. $-27 + 8 = \mathbf{-19}$
2. $-18 - 13 = \mathbf{-31}$
3. $-28 - 1 = \mathbf{-29}$
4. $-24 + 17 = \mathbf{-7}$
5. $-7 - 12 = \mathbf{-19}$
6. $-11 + 17 = \mathbf{6}$
7. $-21 + 15 = \mathbf{-6}$
8. $17 - 15 = \mathbf{2}$
9. $-16 - 17 = \mathbf{-33}$
10. $-29 + 7 = \mathbf{-22}$
11. $-11 - 23 = \mathbf{-34}$
12. $6 - 30 = \mathbf{-24}$
13. $-5 + 2 = \mathbf{-3}$
14. $2 - 11 = \mathbf{-9}$
15. $11 - 26 = \mathbf{-15}$

16. $-16 - 16 = \mathbf{-32}$

17. $6 - 23 = \mathbf{-17}$

18. $21 - 22 = \mathbf{-1}$

19. $-28 - 13 = \mathbf{-41}$

20. $29 - 22 = \mathbf{7}$

21. $5 - 16 = \mathbf{-11}$

22. $-24 - 11 = \mathbf{-35}$

23. $-7 - 5 = \mathbf{-12}$

24. $-13 - 26 = \mathbf{-39}$

25. $-1 - 12 = \mathbf{-13}$

26. $-3 - 5 = \mathbf{-8}$

27. $-21 + 8 = \mathbf{-13}$

28. $-16 + 12 = \mathbf{-4}$

29. $-6 - 1 = \mathbf{-7}$

30. $-3 + 7 = \mathbf{4}$

2.4 Aufgaben zu Kapitel 2.6 bis 2.8

I. Berechne die folgenden Summen/Differenzen:

1. $-1 + 5 - 3 = 4 - 3 = \mathbf{1}$

2. $1 - 5 - 3 = -4 - 3 = \mathbf{-7}$

3. $1 + 5 - 3 = 6 - 3 = \mathbf{3}$

4. $-1 - 5 - 3 = -6 - 3 \mathbf{-9}$

5. $-1 + 5 + 3 = 4 + 3 = \mathbf{7}$

6. $1 - 5 + 3 = -4 + 3 = \mathbf{-1}$

7. $1 + 5 + 3 = 6 + 3 = \mathbf{9}$

8. $-1 - 5 + 3 = -6 + 3 = \mathbf{-3}$

9. $-2 + 7 + 4 = 5 + 4 = \mathbf{9}$

10. $-4 + 2 - 7 = -2 - 7 = \mathbf{-9}$

11. $-7 + 4 - 2 = -3 - 2 = \mathbf{-5}$

12. $-5 - 6 - 2 = -11 - 2 = \mathbf{-13}$

13. $2 - 6 - 3 = -4 - 3 = \mathbf{-7}$

14. $-4-1+4=-5+4=\mathbf{-1}$

15. $7+3-2=10-2=\mathbf{8}$

16. $-2+7+1=5+1=\mathbf{6}$

17. $1-4+5=-3+5=\mathbf{2}$

18. $-7+2-9=-5-9=\mathbf{-14}$

19. $-2-5-6=-7-6=\mathbf{-13}$

20. $1-5-8=-4-8=\mathbf{-12}$

21. $8-5+7-20=3+7-20=10-20=\mathbf{-10}$

22. $-9-4+11-17=-13+11-17=-2-17=\mathbf{-19}$

II. Berechne die folgenden Summen/Differenzen:

1. $-16-6+19=-22+19=\mathbf{-3}$

2. $-5+15+2-13+9=10+2-13+9=12-13+9=-1+9=\mathbf{8}$

3. $-4-10-18-13+8=-14-18-13+8=-32-13+8=-45+8=\mathbf{-37}$

4. $-18-6-7-9+15=-24-7-9+15=-31-9+15=-40+15=\mathbf{-25}$

5. $-17+14-9=-3-9=\mathbf{-12}$

6. $-20-1+3=-21+3=\mathbf{-18}$

7. $16-12+13-29=4+13-29=17-29=\mathbf{-12}$

8. $-2-29+6=-31+6=\mathbf{-25}$

9. $-17+14-9=-3-9=\mathbf{-12}$

10. $-10-3-27=-13-27=\mathbf{-40}$

11. $-19-13+5-1-24=-32+5-1-24=-27-1-24=-28-24=\mathbf{-52}$

12. $-21-2+11=-23+11=\mathbf{-12}$

13. $-25-4+13=-29+13=\mathbf{-16}$

14. $-1-17+20-14=-18+20-14=2-14=\mathbf{-12}$

15. $12+27-3=39-3=\mathbf{36}$

16. $23-26-14+8+27=-3-14+8+27=-17+8+27=-9+27=\mathbf{18}$

17. $26-24+16-5-10=2+16-5-10=18-5-10=13-10=\mathbf{3}$

18. $-6+13-11=7-11=\mathbf{-4}$

19. $17-7+8=10+8=\mathbf{18}$

20. $-21+3+5+16+10=-18+5+16+10=-13+16+10=3+10=\mathbf{13}$

21. $14 - 1 + 10 - 22 - 1 = 13 + 10 - 22 - 1 = 23 - 22 - 1 = 1 - 1 = \mathbf{0}$
22. $-6 - 11 + 17 - 12 + 13 = -17 + 17 - 12 + 13 = -12 + 13 = \mathbf{1}$
23. $6 + 29 - 13 = 35 - 13 = \mathbf{22}$
24. $-16 - 23 - 25 = -39 - 25 = \mathbf{-64}$
25. $-18 + 9 - 20 = -9 - 20 = \mathbf{-29}$
26. $-13 + 22 + 7 + 21 = 9 + 7 + 21 = 16 + 21 = \mathbf{37}$
27. $1 + 10 - 22 + 6 - 11 = 11 - 22 + 6 - 11 = -11 + 6 - 11 = -5 - 11 = \mathbf{-16}$
28. $16 - 17 + 25 = -1 + 25 = \mathbf{24}$
29. $-5 - 29 - 8 - 23 = -34 - 8 - 23 = -42 - 23 = \mathbf{-65}$
30. $-13 + 4 - 12 - 7 = -9 - 12 - 7 = -21 - 7 = \mathbf{-28}$

III. Berechne:

1. $8 - (9 - 4) = 8 - 5 = \mathbf{3}$
2. $13 - (3 - 7) = 13 - (-4) = 13 + 4 = \mathbf{17}$
3. $12 - (-2 - 6) = 12 - (-8) = 12 + 8 = \mathbf{20}$
4. $7 - (-4 + 6) = 7 - 2 = \mathbf{5}$
5. $1 + (3 - 7) = 1 + (-4) = 1 - 4 = \mathbf{-3}$
6. $4 + (-3 - 7) = 4 + (-10) = 4 - 10 = \mathbf{-6}$
7. $0 - (8 - 3) = 0 - 5 = \mathbf{-5}$
8. $0 - (-4 - 9) = 0 - (-13) = \mathbf{13}$
9. $-(-2 + 6) = \mathbf{-4}$
10. $-(4 + 6) = \mathbf{-10}$
11. $-(5 - 12) = -(-7) = \mathbf{7}$
12. $-(-3 - 7) - (8 - 5) = -(-10) - 3 = 10 - 3 = \mathbf{7}$
13. $6 - (8 - 5) - 3 = 6 - 3 - 3 = 3 - 3 = \mathbf{0}$
14. $-(4 - 7) - (-3 - 2) = -(-3) - (-5) = 3 + 5 = \mathbf{8}$
15. $-(-3 + 7) + (8 - 5) + 3 = -4 + 3 + 3 = -1 + 3 = \mathbf{2}$
16. $(4 - 8) - (9 - 6) = -4 - 3 = \mathbf{-7}$
17. $-(-3 + 11) + (8 - 5) = -8 + 3 = \mathbf{-5}$
18. $-(-2 - 6) - (13 - 7) = -(-8) - 6 = 8 - 6 = \mathbf{2}$
19. $-(9 + 4) - (-11 + 17) = -13 - 6 = \mathbf{-19}$

20. $-(9-4+11-17) = -(5+11-17) = -(16-17) = -(-1) = \mathbf{1}$

21. $7-(4+6) = 7-10 = \mathbf{-3}$

22. $-(2-(6-8)) = -(2-(-2)) = -(2+2) = \mathbf{-4}$

IV. Schreibe übersichtlich und berechne:

1. $3-(7-(10-4)) = 3-(7-6) = 3-1 = \mathbf{2}$
2. $14-(5-(8-3)) = 14-(5-5) = 14-0 = \mathbf{14}$
3. $12-(-4-(-2-6)) = 12-(-4-(-8)) = 12-(-4+8) = 12-4 = \mathbf{8}$
4. $3-(-(-4+6)+8) = 3-(-2+8) = 3-6 = \mathbf{-3}$
5. $9-(-(3-7)-(-5+13)) = 9-(-(-4)-8) = 9-(4-8) = 9-(-4) = 9+4 = \mathbf{13}$
6. $-(4-(6-7))+7 = -(4-(-1))+7 = -(4+1)+7 = -5+7 = \mathbf{2}$
7. $(-4-9)-(8-3) = -13-5 = \mathbf{-18}$
8. $-(-5+3)+(7-18) = -(-2)+(-11) = 2-11 = \mathbf{-9}$
9. $-((-5+3)+(7-18)) = -(-2+(-11)) = -(-2-11) = -(-13) = \mathbf{13}$
10. $3-(4-(5-(-7+8))) = 3-(4-(5-1)) = 3-(4-4) = 3-0 = \mathbf{3}$
11. $(-8+15)-(5-(6-18)) = 7-(5-(-12)) = 7-(5+12) = 7-17 = \mathbf{-10}$
12. $-((-4-9)-(12-7)) = -(-13-5) = -(-18) = \mathbf{18}$
13. $6-((8-5)-3) = 6-(3-3) = 6-0 = \mathbf{6}$
14. $1+((-(4-7)+3)-(-1-2)) = 1+((-(-3)+3)-(-1-2)) = 1+((3+3)-(-3)) =$
    $1+(6+3) = 1+9 = \mathbf{10}$
15. $(-(-5))+(-5) = 5-5 = \mathbf{0}$
16. $4-(-(-(-1))) = 4-(-1) = 4+1 = \mathbf{5}$
17. $5+(4-(-3+11)) = 5+(4-8) = 5+(-4) = 5-4 = \mathbf{1}$
18. $4-(-3-9)-(12-25) = 4-(-12)-(-13) = 4+12+13 = 16+13 = \mathbf{29}$
19. $5-((-3+2)-(7+(-9))) = 5-(-1-(7-9)) = 5-(-1-(-2)) = 5-(-1+2) = 5-1 = \mathbf{4}$
20. $1-(2-3+(4-(5-6+(7-8)))) = 1-(-1+(4-(-1+(-1)))) = 1-(-1+(4-(-1-1))) =$
    $1-(-1+(4-(-2))) = 1-(-1+6)) = 1-5 = 1+3 = \mathbf{-4}$

2.5 Aufgaben zu Kapitel 2.9 bis 2.10

I. Berechne:

a) $3 \cdot (-5) = \mathbf{-15}$

b) $(-4) \cdot 8 = \mathbf{-32}$

c) $(-4) \cdot (-4) = \mathbf{16}$

d) $-(-2) \cdot (-3) = 2 \cdot (-3) \mathbf{-6}$

e) $(-5) \cdot (-6) = \mathbf{30}$

f) $8 \cdot (-4) + (-2) \cdot (-3) = -32 + 6 = \mathbf{-26}$

g) $(-3) \cdot 5 + (-7) \cdot (-4) = -15 + 28 = \mathbf{13}$

h) $(-3) \cdot (-2) + 2 \cdot (-1) + 5 = 6 + (-2) + 5 = 4 + 5 = \mathbf{9}$

i) $(-5) \cdot 3 + 9 + (-4) \cdot (-2) = -15 + 9 + 8 = -6 + 8 = \mathbf{2}$

j) $(-3) \cdot 8 - (-7) \cdot (-5) = -24 - 35 = \mathbf{-59}$

k) $(-1) \cdot (-1) = \mathbf{1}$

l) $(-1) \cdot 1 = \mathbf{-1}$

m) $1 \cdot (-1) = \mathbf{-1}$

n) $(-1) \cdot 8 + (-1) \cdot 5 = -8 - 5 = \mathbf{-13}$

o) $(-1) \cdot 9 + (-1) \cdot 4 + 7 + (-1) \cdot 3 = -9 - 4 + 7 - 3 = -13 + 7 - 3 = -6 - 3 = \mathbf{-9}$

II. Berechne:

1. $54 : (-3) = \mathbf{-18}$
2. $(-48) : 8 = \mathbf{-6}$
3. $(-56) : (-7) = \mathbf{8}$
4. $(-144) : 6 = \mathbf{-24}$
5. $(-54) : (-6) = \mathbf{9}$
6. $\dfrac{-3}{4} = -\dfrac{\mathbf{3}}{\mathbf{4}}$
7. $\dfrac{10}{-16} = -\dfrac{\mathbf{5}}{\mathbf{8}}$
8. $\dfrac{-10}{-12} = \dfrac{\mathbf{5}}{\mathbf{6}}$
9. $\dfrac{-2 + 3 - 7 + 4}{5 - 8 + 2 - 6 + 1} = \dfrac{1 - 3}{-3 - 4 + 1} = \dfrac{-2}{-6} = \dfrac{\mathbf{1}}{\mathbf{3}}$

10. $\dfrac{3-8+1-11+5}{5-8+2-6+1-(-10)} = \dfrac{-5-10+5}{-3-4+11} = \dfrac{-15+5}{-7+11} = \dfrac{-10}{4} = -\mathbf{\dfrac{5}{2}}$

11. $\dfrac{1+4-1+4-2+1-3+5}{1-7+3-20+50-8} = \dfrac{5+3-1+2}{-6-17+42} = \dfrac{8-3}{-23+42} = \mathbf{\dfrac{5}{19}}$

12. $\dfrac{-3-1+4+1-5+9-2-7}{2+2-3-6-0+6-8} = \dfrac{-4+5+4-9}{4-9-2} = \dfrac{1-5}{-5-2} = \dfrac{-4}{-7} = \mathbf{\dfrac{4}{7}}$

13. $\dfrac{-4-7-7+1-2+1-3}{6-9-3+1-4+7-2} = \dfrac{-11-6-1-2}{-3-2+3} = \dfrac{-17-3}{-5+3} = \dfrac{-21}{-2} = \mathbf{\dfrac{21}{2}}$

14. $\dfrac{3-8+1-11+5}{5-8+2-6+1-(-10)} = \dfrac{-5-10+5}{-3-4+11} = \dfrac{-15+5}{-7+11} = \dfrac{-10}{4} = -\mathbf{\dfrac{5}{2}}$

15. $\dfrac{2-7+1-8+2-8+1-8}{3-1-6-2-2+7+7-7} = \dfrac{-5-7-6-7}{2-8+5} = \dfrac{-12-13}{-6+5} = \dfrac{-25}{-1} = \mathbf{25}$

16. $\dfrac{-3}{4} + \dfrac{2}{-5} = -\dfrac{3}{4} - \dfrac{2}{5} = -\dfrac{15}{20} - \dfrac{8}{20} = -\mathbf{\dfrac{23}{20}}$

17. $\dfrac{-4}{-5} + \dfrac{2}{-8} = \dfrac{4}{5} - \dfrac{1}{4} = \dfrac{16}{20} - \dfrac{5}{20} = \mathbf{\dfrac{11}{20}}$

18. $\dfrac{4}{-7} - \dfrac{-3}{5} = -\dfrac{4}{7} + \dfrac{3}{5} = -\dfrac{20}{35} + \dfrac{21}{35} = \mathbf{\dfrac{1}{35}}$

19. $\dfrac{-1}{9} - \dfrac{1}{-11} = -\dfrac{1}{9} + \dfrac{1}{11} = -\dfrac{11}{99} + \dfrac{9}{99} = -\mathbf{\dfrac{2}{99}}$

20. $\dfrac{5}{-13} + \dfrac{4}{-5} = -\dfrac{5}{13} - \dfrac{4}{5} = -\dfrac{25}{65} - \dfrac{52}{65} = -\mathbf{\dfrac{77}{65}}$

2.6 Aufgaben zu Kapitel 2.11

I. Berechne:

a) $2^{-5} = \dfrac{1}{2^5} = \dfrac{1}{2^3} \cdot \dfrac{1}{2^2} = \dfrac{1}{8} \cdot \dfrac{1}{4} = \mathbf{\dfrac{1}{32}}$

b) $(-3)^{-4} = -\dfrac{1}{3^4} = -\dfrac{1}{3^2} \cdot \dfrac{1}{3^2} = -\dfrac{1}{9} \cdot \dfrac{1}{9} = -\dfrac{1}{81}$

c) $(-3)^{-3} = -\dfrac{1}{3^3} = -\dfrac{1}{3^2} \cdot \dfrac{1}{3^1} = -\dfrac{1}{9} \cdot \dfrac{1}{3} = -\mathbf{\dfrac{1}{27}}$

d) $5^{-4} = \dfrac{1}{5^4} = \dfrac{1}{5^2} \cdot \dfrac{1}{5^2} = \dfrac{1}{25} \cdot \dfrac{1}{25} = \dfrac{1}{125} \cdot \dfrac{1}{5} = \mathbf{\dfrac{1}{625}}$

e) $2^{-10} = \dfrac{1}{2^{10}} = \dfrac{1}{2^5} \cdot \dfrac{1}{2^5} = \dfrac{1}{32} \cdot \dfrac{1}{32} = \dfrac{1}{64} \cdot \dfrac{1}{16} = \dfrac{1}{128} \cdot \dfrac{1}{8} = \dfrac{1}{256} \cdot \dfrac{1}{4} = \dfrac{1}{512} \cdot \dfrac{1}{2} = \mathbf{\dfrac{1}{1024}}$

f) $10^{-3} = \dfrac{1}{10^3} = \mathbf{\dfrac{1}{1000}}$

g) $2^9 \cdot 2^8 \cdot 2^{-14} = 2^{9+8-14} = 2^{17-14} = 2^3 = \mathbf{8}$

h) $2^7 \cdot 3^7 \cdot 2^{-5} \cdot 3^{-5} = 2^{7-5} \cdot 3^{7-5} = 2^2 \cdot 3^2 = 4 \cdot 9 = \mathbf{36}$

i) $2^{93} \cdot 2^{27} \cdot 2^{-42} \cdot 2^{-69} = 2^{93+27-42-69} = 2^{120-111} = 2^9 = 2^5 \cdot 2^4 = 32 \cdot 16 = 64 \cdot 8 = 128 \cdot 4 = 256 \cdot 2 = \mathbf{512}$

j) $2^{23} \cdot 3^2 \cdot 2^{-3} \cdot 2^{-16} = 2^{23-3-16} \cdot 3^2 = 2^4 \cdot 3^2 = 16 \cdot 9 = \mathbf{144}$

k) $2^5 + 2^5 \cdot 3^5 \cdot 3^{-5} = 32 + 32 \cdot 3^{5-5} = 32 + 32 \cdot 3^0 = 32 + 32 \cdot 1 = \mathbf{64}$

l) $2^{-3} \cdot 3^2 = \dfrac{3^2}{2^3} = \dfrac{\mathbf{9}}{\mathbf{8}}$

II. Berechne:

a) $\dfrac{3^5 \cdot 3^8}{3^2 \cdot 3^4} = \dfrac{3^{5+8}}{3^{2+4}} = \dfrac{3^{13}}{3^6} = 3^{13-6} = 3^7 = 3^4 \cdot 3^3 = 81 \cdot 27 = 243 \cdot 9 = \mathbf{2.187}$

b) $\dfrac{2^7 \cdot 2^4}{2^5} = \dfrac{2^{7+4}}{2^5} = \dfrac{2^{11}}{2^5} = 2^{11-5} = 2^6 = 2^3 \cdot 2^3 = 8 \cdot 8 = \mathbf{64}$

c) $\dfrac{2^5 \cdot 3^5}{2^3 \cdot 3^2} = 2^{5-3} \cdot 3^{5-2} = 2^2 \cdot 3^3 = 4 \cdot 27 = \mathbf{108}$

d) $\dfrac{(2^3)^2}{2^2 \cdot 2^3} = \dfrac{2^{3\cdot 2}}{2^{2+3}} = \dfrac{2^6}{2^5} = 2^{6-5} = 2^1 = \mathbf{2}$

e) $\dfrac{(2^2)^2 \cdot (3^2)^2}{2^2 \cdot 3^2} = 2^{2\cdot 2-2} \cdot 3^{2\cdot 2-2} = 2^{4-2} \cdot 3^{4-2} = 2^2 \cdot 3^2 = 4 \cdot 9 = \mathbf{36}$

f) $\dfrac{2^4 \cdot 3^4 \cdot 2^5}{3^3 \cdot (3^2)^2} = 2^{4+5} \cdot 3^{4-3-2\cdot 2} = 2^9 \cdot 3^{1-4} = 2^9 \cdot 3^{-3} = \dfrac{2^9}{3^3} = \dfrac{\mathbf{512}}{\mathbf{27}}$

2.7 Aufgaben zu Kapitel 2.12

I. Berechne:

a) $3,517 \cdot 10^3 = \mathbf{3.517}$

b) $1,4142135 \cdot 10^{-3} = \mathbf{0,001.414.213.5}$

c) $1,73205 \cdot 10^{-1} = \mathbf{0,173.205}$

d) $3,14159 \cdot 10^{12} = \mathbf{3.141.590.000.000}$

e) $2,7182818 \cdot 10^{-7} = \mathbf{0,000.000.271.828.18}$

f) $3,1622776 \cdot 10^{-1} = \mathbf{0,316.227.76}$

g) $8660,254 \cdot 10^{-4} = \mathbf{0,866.025.4}$

h) $7071067,8 \cdot 10^{-8} = \mathbf{0,070.710.678}$

i) $0,30102999 \cdot 10^{-6} = \mathbf{0,000.000.301.029.99}$

j) $693,147180 \cdot 10^7 = \mathbf{6.931.471.800}$

II. Schreibe in wissenschaftlicher Notation:

a) $7.325 = \mathbf{7,325 \cdot 10^3}$

b) $1.358.732 = \mathbf{1,358.732 \cdot 10^6}$

c) $972.144 = \mathbf{9,721.44 \cdot 10^5}$

d) $0,141.537 = \mathbf{1,415.37 \cdot 10^{-1}}$

e) $0,053.428.9 = \mathbf{5,342.89 \cdot 10^{-2}}$

f) $0,000.024.698 = \mathbf{2,469.8 \cdot 10^{-5}}$

g) $14.031.879 = \mathbf{1,403.187.9 \cdot 10^7}$

h) $0,001.710.927 = \mathbf{1,710.927 \cdot 10^{-3}}$

i) $0,000.000.000.08 = \mathbf{8,0 \cdot 10^{-11}}$

j) $432.177.605 = \mathbf{4,321.776.05 \cdot 10^8}$

3 Buchstabenrechnung

3.1 Aufagaben zu Kapitel 3.1

I. Berechne:

a) $8a - 5a = \mathbf{3a}$

b) $-5a + 8a = \mathbf{3a}$

c) $-7a + 3a = \mathbf{-4a}$

d) $13a - 19a = \mathbf{-6a}$

e) $-21a + 8a = \mathbf{-13a}$

f) $-8b - 5b = \mathbf{-13b}$

g) $8b - 5b + 7b - 20b = 3b - 13b = \mathbf{-10b}$

h) $-9c - 4c + 11c - 17c = -13c - 6c = \mathbf{-19c}$

i) $-(9a + 4a) - (-11a + 17a) = -13a - 6a = \mathbf{-19a}$

j) $-(9a - 4a + 11a - 17a) = -(5a - 6a) = -(-a) = \mathbf{a}$

k) $4x - 9x = \mathbf{-5x}$

l) $-5x + 18x = \mathbf{13x}$

m) $12x - 7x = \mathbf{5x}$

n) $8x - 5y - 3x + 2y = \mathbf{5x - 3y}$

o) $x - 6y + 4y - 3x = \mathbf{-2x - 2y}$

p) $8a - 5b + 2c - 3a + 7b - 4c = \mathbf{5a + 2b - 2c}$

II. Berechne:

a) $(4a+5b)+(2a+3b)=4a+5b+2a+3b=\mathbf{6a+8b}$

b) $(4a+5b)+(2a-3b)=4a+5b+2a-3b=\mathbf{6a+2b}$

c) $(4a+5b)-(2a-3b)=4a+5b-2a+3b=\mathbf{2a+8b}$

d) $(4a+5b)-(-2a+3b)=4a+5b+2a-3b=\mathbf{6a+2b}$

e) $(4a+5b)-(-2a-3b)=4a+5b+2a+3b=\mathbf{6a+8b}$

f) $-(9a-4b)-(-11a+17b)=-9a+4b+11a-17b=\mathbf{2a-13b}$

g) $7a-(4b+6a)=7a-4b-6a=\mathbf{a-4b}$

h) $-(2a-(6a-8b))=-(2a-6a+8b)=-2a+6a-8b=\mathbf{4a-8b}$

i) $5a-((-3b+2a)-(7a+(-9b)))=5a-(-3b+2a-7a-(-9b))=5a-(6b-5a)=5a-6b+5a=\mathbf{10a-6b}$

j) $x-[2y-3x+(4x-(5y-6x+(7x-8y)))]=x-2y+3x-(4x-5y+6x-(7x-8y))=4x-2y-(10x-5y-7x+8y)=4x-2y-(3x+3y)=4x-2y-3x-3y=\mathbf{x-5y}$

III. Berechne:

a) $6c-3c=\mathbf{3c}$

b) $7d-11d=\mathbf{-4d}$

c) $-5x+8x=\mathbf{3x}$

d) $-4y-7y=\mathbf{-11y}$

e) $5c-2b+4a+1-3b+2a-7c-3=\mathbf{6a-5b-2c-2}$

f) $-8y+2z-x-4z+2y+5x=\mathbf{4x-6y-2z}$

g) $4w-2v+8u-5w-11u=\mathbf{-3u-2v-w}$

h) $3n-2m+2n-4n-9n=(3+2-4-9)n-2m=(5-13)n-2m=\mathbf{-2m-8n}$

IV. Berechne:

a) $2a-3b+(4b-a)-(7a-2b-3b)=2a-3b+4b-a-(7a-5b)=2a+b-a-7a+5b=(2-1-7)a+(-1+5)b=\mathbf{-6a+4b}$

b) $2b-3c+4a-(-6c+2a-2c+3b-7a)=2b-3c+4a-(-8c-5a+3b)=2b-3c+4a+8c+5a-3b=\mathbf{9a-1b+5c}$

c) $-(-2x+7y-3x)+(5y-6x+2x-3y)-3x+9y=-(-5x+7y)+(2y-4x)-3x+9y=5x-7y+2y-4x-3x+9y=(5-4-3)x+(-7+2+9)y=\mathbf{-2x+4y}$

d) $-3y-4x-(2y-x+3y)=-3y-4x-(5y-x)=-3y-4x-5y+x=(-4+1)x+(-3-5)y=\mathbf{-3x-8y}$

e) $6-3b+2a-(2b-3a-(4a-3b))=6-3b+2a-(2b-3a-4a+3b)=6-3b+2a-(5b-7a)=6-3b+2a-5b+7a=(2+7)a+(-3-5)b+6=\mathbf{9a-8b+6}$

f) $-7x+3-(3x+1-7x+3) = -7x+3-(-4x+4) = -7x+3+4x-4 = \mathbf{-3x-1}$

g) $2-(3v-5w)+(2w-4-v+5w+1) = 2-3v+5w+(7w-3-v) = 2-3v+5w+(7w-3-v) = 2-3v+5w+7w-3-v = (-3-1)v+(5+7)w-1 = \mathbf{-4v+12w-1}$

h) $(2-(u-3v)+5u)-(3v+1-(-2v+3u)) = (2-u+3v+5u)-(3v+1+2v-3u) = (2+4u+3v)-(5v+1-3u) = 2+4u+3v-5v-1+3u = (4+3)u+(3-5)v+1 = \mathbf{7u-2v+1}$

3.2 Aufgaben zu Kapitel 3.2 bis 3.4

I. Multipliziere aus:

a) $x\cdot(x+5) = \mathbf{x^2+5x}$

b) $(x+3)\cdot x = \mathbf{x^2+3x}$

c) $x\cdot(x-7) = \mathbf{x^2-7x}$

d) $a\cdot(8-a) = \mathbf{-a^2+8a}$

e) $-3\cdot(x-12) = \mathbf{-3x+36}$

f) $a\cdot(-x-5) = \mathbf{-ax-5a}$

g) $-a\cdot(-x+5) = \mathbf{ax-5a}$

h) $x\cdot(8x-5z-3+2y) = \mathbf{8x^2+2xy-5xz-3x}$

i) $(a-2b+3c)\cdot a = \mathbf{a^2-2ab+3ac}$

j) $(a-b+c)\cdot(-a) = \mathbf{-a^2+ab-ac}$

k) $(a+3)\cdot(b+4) = \mathbf{ab+4a+3b+12}$

l) $(a+2)\cdot(-b+5) = \mathbf{-ab+5a-2b+10}$

m) $(x+6)\cdot(x+3) = x^2+3x+6x+18 = \mathbf{x^2+9x+18}$

n) $(a+1)\cdot(b+2) = \mathbf{ab+2a+b+2}$

o) $(2x+3)\cdot(5x-7) = 10x^2-14x+15x-21 = \mathbf{10x^2+x-21}$

p) $(2x+3)\cdot(3x-5) = 6x^2-10x+9x-15 = \mathbf{6x^2-x-15}$

II. Multipliziere aus:

a) $(x+1)\cdot(x+2) = x^2+2x+x+2 = \mathbf{x^2+3x+2}$

b) $(x+4)\cdot(x+5) = x^2+5x+4x+20 = \mathbf{x^2+9x+20}$

c) $(x+2)\cdot(x-3) = x^2-3x+2x-6 = \mathbf{x^2-x-6}$

d) $(x-3)\cdot(x+4) = x^2+4x-3x-12 = \mathbf{x^2+x-12}$

e) $(x-5)\cdot(x-1) = x^2-x-5x+5 = \mathbf{x^2-6x+5}$

f) $(-x+3)\cdot(x-4) = -x^2+4x+3x-12 = \mathbf{-x^2+7x-12}$

g) $(-x-1)\cdot(-x+2) = x^2-2x+x-2 = \mathbf{x^2-x-2}$

h) $(-x-3)\cdot(-x-7) = x^2+7x+3x+21 = \mathbf{x^2+10x+21}$

i) $(a-2)\cdot(b+5) = \mathbf{ab+5a-2b-10}$

j) $(u+1)\cdot(-v+6) = \mathbf{-uv+6u-v+6}$

k) $(x-4)\cdot(b-3) = \mathbf{bx-4b-3x+12}$

l) $(z+2)\cdot(w-1) = \mathbf{wz+2w-z-2}$

m) $(x-7)\cdot(y+6) = \mathbf{xy+6x-7y-42}$

n) $(k-2)\cdot(k-3) = k^2-3k-2k+6 = \mathbf{k^2-5k+6}$

o) $(m+4)\cdot(n-4) = \mathbf{mn-4m+4n-16}$

p) $(a+3)\cdot(a+3) = \mathbf{a^2+6a+9}$

III. Multipliziere aus:

a) $(3x+2)\cdot(4x+5) = 12x^2+15x+8x+10 = \mathbf{12x^2+23x+10}$

b) $(4x+3)\cdot(2x+1) = 8x^2+4x+6x+3 = \mathbf{8x^2+10x+3}$

c) $(5x+2)\cdot(3x-4) = 15x^2-20x+6x-8 = \mathbf{15x^2-14x-8}$

d) $(2x-3)\cdot(y+4) = \mathbf{2xy+4x-3y-12}$

e) $(3y-5)\cdot(2x-6) = \mathbf{6xy-10x-18y+30}$

f) $(-2x+3y)\cdot(5x-4y) = -10x^2+8xy+15xy-12y^2 = \mathbf{-10x^2+23xy-12^2}$

g) $(-x-y)\cdot(-2x+y) = 2x^2-xy+2xy-y^2 = \mathbf{2x^2+xy-y^2}$

h) $(-a-2b)\cdot(-a-6b) = a^2+6ab+2ab+12b^2 = \mathbf{a^2+8ab+12b^2}$

i) $(3a-2b)\cdot(5a+b) = 15a^2+3ab-10ab-2b^2 = \mathbf{15a^2-7ab-2b^2}$

j) $(u+v)\cdot(-u+4v) = -u^2+4uv-uv+4v^2 = \mathbf{-u^2+3uv+4v^2}$

k) $(2u-4w)\cdot(3v-5w) = \mathbf{6uv-10uw-12vw+20w^2}$

l) $(w+2)\cdot(z-1) = \mathbf{wz-w+2z-2}$

m) $(5x-6y)\cdot(2x+3y) = 10x^2+15xy-12xy-18y^2 = \mathbf{10x^2+3xy-18y^2}$

n) $(a-2b+3c)\cdot(2a-b+5) = 2a^2-ab+5a-4ab+2b^2-10b+6ac-3bc+15c = \mathbf{2a^2-5ab+6ac+2b^2-3bc+5a-10b+15c}$

o) $(2m+3n)\cdot(3m-2n) = 6m^2-4mn+9mn-6n^2 = \mathbf{6m^2+5mn-6n^2}$

p) $(2a+3)\cdot(2a-3) = \mathbf{4a^2-9}$

IV. Multipliziere aus:

a) $\left(\frac{3}{2}x + \frac{1}{2}y\right) \cdot \left(\frac{5}{2}x - \frac{1}{2}y\right) = \frac{1}{2} \cdot (3x + y) \cdot (5x - y) = \frac{1}{2} \cdot (15x^2 - 3xy + 5xy - y^2) = \frac{1}{2} \cdot (15x^2 + 2xy - y^2) = \mathbf{\frac{15}{2}x^2 + xy - \frac{1}{2}y^2}$

b) $\left(-\frac{2}{5}x + \frac{1}{3}y\right) \cdot \left(\frac{3}{4}x - \frac{5}{6}y\right) = -\frac{6}{20}x^2 + \frac{10}{30}xy + \frac{3}{12}xy - \frac{5}{18} = -\frac{3}{10}x^2 + \frac{1}{3}xy + \frac{1}{4}xy - \frac{5}{18} = \mathbf{-\frac{3}{10}x^2 + \frac{7}{12}xy - \frac{5}{18}y^2}$

c) $x \cdot \left(1 + \frac{1}{2}x \cdot \left(1 + \frac{1}{3}x\right)\right) = x \cdot \left(1 + \frac{1}{2}x + \frac{1}{6}x^2\right) = \mathbf{\frac{1}{6}x^3 + \frac{1}{2}x^2 + x}$

d) $(0,2x - 1,3y) \cdot (0,5x + 4y) = 0,01x^2 + 0,8xy - 6,5xy - 5,2y^2 = \mathbf{0,01x^2 - 5,7xy - 5,2y^2}$

e) $(x - 1) \cdot (1 + x + x^2 + x^3) = x + x^2 + x^3 + x^4 - 1 - x - x^2 - x^3 = \mathbf{x^4 - 1}$